

Painter by Numbers: Multi-Class Artist Classification

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I. INTRODUCTION

Artist identification from paintings is a challenging fine-grained classification problem requiring discrimination between subtle stylistic differences. This project develops multi-class classifiers to predict the artist among 24 painters using the Painter by Numbers dataset. We systematically explore custom CNN architectures, transfer learning approaches, and traditional ML methods to understand optimal design choices for artistic style recognition.

II. DATASET AND PREPROCESSING

The dataset contains 11,880 images from 24 artists (495 images each, balanced sampling). Images were resized to 224×224 with ImageNet normalization for CNNs. For SVMs, we extracted HOG features (6,084-D), color histograms (96-D), and ResNet-18 deep features (512-D), all standardized before training. Ten corrupted images were replaced, and stratified sampling ensured balanced train/validation/test splits.

III. METHODOLOGY

A. Custom CNN Experiments

We systematically evaluated CNN design choices through three experimental groups: **Exp1-4 (Activation Study)** comparing ReLU, LeakyReLU, ELU variants with 4-block architectures; **Exp5 (Modern Activation)** evaluating GELU; **Exp6-7 (Depth Study)** exploring deeper (5-block) and shallower (3-block) architectures. Additionally, Exp3 examined AvgPool versus MaxPool. All models used Conv2D→BatchNorm→Activation→Pooling blocks with [64,128,256,512,512] channels, global average pooling, dropout (0.5), and dense layers [512,256]. Training used AdamW optimizer, label smoothing cross-entropy, ReduceLROnPlateau scheduler, and early stopping (patience=15).

B. Transfer Learning

Pretrained models: (1) ResNet-18 with frozen early layers and custom head, (2) EfficientNet-B0 with partial freezing and dropout, (3) CNN Ensemble combining both via soft-voting on logits.

C. Traditional Machine Learning

SVMs with: (1) HOG+RBF, (2) Color+One-vs-Rest, (3) ResNet features+RBF (5-fold CV, C=10), (4) Fused Features (ResNet+HOG+Color)+Linear SVM.

TABLE I
MODEL PERFORMANCE COMPARISON

Model/Experiment	Test Acc (%)
<i>Custom CNN Experiments</i>	
Exp5: GELU+MaxPool-4Blocks	68.35
Exp7: ReLU+MaxPool-3Blocks	67.85
Exp2: LeakyReLU+MaxPool-4Blocks	66.78
Exp1: ReLU+MaxPool-4Blocks	66.16
Exp4: ELU+MaxPool-4Blocks	58.53
Exp6: ReLU+MaxPool-5Blocks	58.25
Exp3: ReLU+AvgPool-4Blocks	56.40
<i>Transfer Learning</i>	
CNN Ensemble (ResNet+EfficientNet)	81.87
ResNet-18 Transfer Learning	78.90
EfficientNet-B0 Transfer Learning	75.98
<i>Traditional ML</i>	
Fused Features + Linear SVM	79.80
ResNet Features + RBF SVM	74.19
Color Histogram + OvR SVM	34.18
HOG + RBF SVM	4.10

IV. RESULTS

Key Findings: (1) *Activation Functions:* GELU outperformed ReLU/LeakyReLU (2% gain); ELU underperformed. (2) *Pooling:* MaxPool significantly superior to AvgPool (66.16% vs 56.40%). (3) *Depth:* 3-4 blocks optimal; 5 blocks showed overfitting. (4) *Per-Artist:* Strong performance on distinctive styles: Beksiński (97.33%), Renoir (98.65%), Repin (97.33%); challenging classes: Chagall (71.62%), impressionists with overlapping palettes.

V. CONCLUSIONS AND CHALLENGES

Conclusions: Transfer learning dramatically outperformed custom CNNs (+13.5% over best custom model). GELU activation and max pooling are optimal for custom architectures. Shallow-to-moderate depth (3-4 blocks) balances performance and efficiency. Fused features (79.80%) provide lightweight alternative to ensembles. Traditional features alone insufficient for artistic discrimination.

Challenges: Computational intensity of training 7 CNN variants plus transfer learning models; managing overfitting in deeper architectures despite regularization; discriminating similar artistic movements (impressionists); CPU-intensive HOG extraction; handling corrupted images and ensuring data quality.

Future Work: Vision Transformers, art-specific pretraining, advanced augmentation strategies, Grad-CAM explainability analysis, ensemble methods combining custom and pretrained models.